

CE 310 — Week 4 Assignment: Poisson Probabilities and Return Periods

Due: Wed Sep 23, 9:00 AM — submit to D2L

Overview

This assignment extends the Week 4 in-class exercise into new Poisson scenarios. You will compute event probabilities at thresholds and time horizons not covered in the exercise, interpret those results through an engineering design lens, and explain in writing what the Poisson model reveals that a simple average cannot.

All three problems use the same dataset: Week4_Tucson_Precipitation_Daily_2013_2022.csv

How to Submit

Read this before you submit. Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	.xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF
Sheet name	Must be exactly Answers — do not rename or delete it
Where answers go	Column A of the Answers sheet labels every row — P1b_written, P2a_written, and so on. Each question names its row; type into the shaded cell on that row
Major declaration	Enter CE or ArcE in cell B4 — this determines which 2b row is graded
Column layout	Enter your formula in Column C and the resulting value in Column D — do not insert, delete, or move columns

Requirement	Detail
Formulas	Type them as text — put an apostrophe before the =, e.g. '=COUNTIF(B2:B8761,">90")', so Excel stores the formula instead of running it. Screenshots are not accepted
Written responses	Type them in the yellow cells lower down the same sheet — do not submit a separate PDF
Name and NetID	Type them in the Name and NetID cells at the top of the Answers sheet — that is what matches your work to your student record. A 5-point deduction applies if the NetID is blank.
File name	CE310_W4_Assignment_<NetID>.xlsx (e.g. CE310_W4_Assignment_abc123.xlsx)
Submission	Upload the .xlsx file to D2L → Assignments by 9:00 AM, Wed Sep 23

Problem 1 — New Poisson Probabilities (30 pts)

The exercise computed λ for Summer, Spring, Winter, and July. This problem asks you to compute three probabilities not found in the exercise — using a new threshold, a new season, and a new time horizon.

Part 1a — An Extreme July (10 pts)

July's $\lambda = 10.3$ rain days per month. Use this rate to compute:

$$P(X \geq 12 \mid \lambda_{\text{july}})$$

The probability that a July will have **12 or more rain days** — roughly a 40% increase over the mean.

Note: Use your exercise B6 answer for λ_{july} . Do not re-derive it from scratch.

Show your formula and the result. In 1–2 sentences, interpret this value: is a 12-rain-day July a rare exception or a reasonably common outcome?

Enter your response in the shaded cell on row P1a_written of the Answers sheet.

Part 1b — A Dry Summer Month (10 pts)

The exercise computed $P(X = 0 \mid \lambda_{\text{spring}})$ and $P(X = 0 \mid \lambda_{\text{winter}})$ but not for Summer. Compute:

$P(X = 0 \mid \lambda_{\text{summer}})$

The probability that a Summer month (June, July, or August) has **zero rain days**.

Note: Use your exercise A7 answer for λ_{summer} .

Show your formula and result. In 2–3 sentences, compare this to the $P(\text{dry spring month})$ from exercise B3. Explain physically why the probability of a dry summer month is so much lower than a dry spring month in Tucson.

Enter your response in the shaded cell on row P1b_written of the Answers sheet.

Part 1c — A Very Dry Spring (10 pts)

Compute:

$P(X \leq 2 \mid \lambda_{\text{spring}})$

The probability that a Spring month has **2 or fewer rain days** — a noticeably dry spring.

Note: Use your exercise B2 answer for λ_{spring} .

Show your formula and result. In 1–2 sentences, explain what the CDF value $P(X \leq 2)$ tells you about the typical spring month in Tucson.

Enter your response in the shaded cell on row P1c_written of the Answers sheet.

Problem 2 — Seasonal Rain Rate and Building Envelope (18 pts)

Your exercise computed four seasonal λ values. This problem uses the Fall rate — a season the exercise did not analyze in depth — and connects it to building envelope performance.

2a — both tracks. Compute Fall lambda from the dataset:

Report the **Fall rain day count** (in the P2a_count row) and λ_{fall} (in the P2a_lambda row). Fall is September, October and November across all 10 years — 30 season-months in total, so $\lambda_{\text{fall}} = (\text{rain days}) \div 30$.

In 1–2 sentences, place Fall in context relative to your four seasonal λ values: which season is wettest, which is driest, and where does Fall rank?

Enter your response in the shaded cell on row P2a_written of the Answers sheet.

2b — answer ONLY the version matching your major. Declare CE or ArcE in cell B4 of your Answer Sheet; that declaration determines which of the two rows below is graded. Leave the other blank.

2b — Track A (Civil Engineering). Municipal drainage infrastructure is specified against a *return period* rather than a monthly rate. A culvert is designed for the **100-year storm** — an event with a 1% chance of occurring in any given year — and the structure is expected to remain in service for a **50-year design life**.

Compute the probability that **at least one** 100-year storm occurs during that 50-year design life:

$$P(\text{at least one}) = 1 - e^{-t/T}$$

where $t = 50$ years (design life) and $T = 100$ years (return period).

Note: In Excel this is $= (1 - \text{EXP}(-50/100)) * 100$. This is the Poisson zero-event complement — the same structure as $P(X = 0)$ in Part 1b, applied to a time horizon instead of a month. Most students expect the answer to be 50%; it is not, and that gap is the point.

CE students: enter this in the shaded cell on row P2b_CE of the Answers sheet.

2b — Track B (Architectural Engineering). Building envelope performance depends on how often moisture exposure repeats within a season, not on a once-a-century design event. For the Fall season, compute:

$P(X \geq 5 \mid \lambda_{\text{fall}})$

The probability that a Fall month has **5 or more rain days** — a level at which prolonged moisture exposure becomes a concern for cladding, flashing, and membrane performance.

Note: Use $= (1 - \text{POISSON.DIST}(4, \lambda_{\text{fall}}, \text{TRUE})) * 100$. Note the 4, not the 5: $P(X \geq 5) = 1 - P(X \leq 4)$.

ArcE students: enter this in the shaded cell on row P2b_ArcE of the Answers sheet.

Show your formula and report the result.

2c. In 3–4 sentences, synthesize **your own 2b result** with the seasonal λ table from Chart 2 of the exercise. Explain why an engineer should use a Poisson exceedance probability rather than simply citing the annual average rain days. Reference ASHRAE 90.1-2022's requirement to size drainage systems for design-storm conditions. CE students: frame the recommendation around sizing a culvert whose service life spans many return periods. ArcE students: frame it around specifying window flashing, roofing membrane, and exterior cladding.

Enter your response in the shaded cell on row P2c_written of the Answers sheet.

Problem 3 — Reflection (12 pts)

Answer in **one paragraph (100–150 words)**:

A building owner reviews your Week 4 in-class exercise and says: “*Tucson gets about 3.9 rain days per month on average, so I’ll size my roof drains for 4 rain days per month. That should cover it.*”

Using at least **one specific number from your exercise or this assignment**, explain why this reasoning is flawed. Your response should address what $P(X > 4 \mid \lambda_{\text{overall}})$ tells you, and what the seasonal Poisson analysis reveals that the annual average obscures.

Enter your response in the shaded cell on row P3_written of the Answers sheet.

Grading Summary

Problem	Points
Problem 1a — $P(X \geq 12 \mid \text{July})$: formula, result, 1–2 sentence interpretation	10
Problem 1b — $P(X = 0 \mid \text{Summer})$: formula, result, 2–3 sentence comparison to spring	10
Problem 1c — $P(X \leq 2 \mid \text{Spring})$: formula, result, 1–2 sentence CDF interpretation	10

Problem	Points
Problem 2 — Seasonal Rain Rate and Building Envelope (2a count + λ_{fall} + written, 2b by major, 2c written)	18
Problem 3 — Reflection paragraph (100–150 words, at least one specific number cited)	12
Total	60 pts